

Solutions to Exercise 1B

Fundamental Operations on Integers: Addition and Subtraction

This guide provides the complete, step-by-step solutions for **Exercise 1B** from the textbook. All steps are grounded in core algebraic properties of the set of integers ($\mathbb{Z}$).

1. Add the following integers

For addition of signed numbers, we apply three simple rules:

1. **Like signs (positive):** Sum their absolute values and prefix with a plus sign.
2. **Like signs (negative):** Sum their absolute values and prefix with a minus sign.
3. **Unlike signs:** Find the difference of their absolute values and prefix with the sign of the integer with the larger absolute value.

(i) **64 and 36**

Both numbers are positive.

$$64 + 36 = \mathbf{100}$$

(ii) **73 and -37**

These have unlike signs. Since $|73| > |-37|$, the result is positive.

$$73 + (-37) = 73 - 37 = \mathbf{36}$$

(iii) **-26 and -45**

Both numbers are negative. Sum their absolute values and apply a negative sign.

$$(-26) + (-45) = -(26 + 45) = \mathbf{-71}$$

(iv) **-51 and 25**

These have unlike signs. Since $|-51| > |25|$, the result is negative.

$$(-51) + 25 = -(51 - 25) = \mathbf{-26}$$

(v) **100 and -32**

Unlike signs. Since $|100| > |-32|$, the result is positive.

$$100 + (-32) = 100 - 32 = \mathbf{68}$$

(vi) **0 and -21**

Zero is the additive identity. Adding zero to any number results in that number.

$$0 + (-21) = \mathbf{-21}$$

2. Find the additive inverse of

Theoretical Rule: Additive Inverse

The **additive inverse** of an integer a is $-a$, which satisfies the property:

$$a + (-a) = (-a) + a = 0$$

Concretely, to find the additive inverse of any integer, we simply change its sign.

(i) **23**

The additive inverse is **-23** because $23 + (-23) = 0$.

(ii) **-33**

The additive inverse is **33** because $-33 + 33 = 0$.

(iii) **-1**

The additive inverse is **1** because $-1 + 1 = 0$.

(iv) **-476**

The additive inverse is **476** because $-476 + 476 = 0$.

3. Evaluate

Subtraction as Addition of Additive Inverse

For any two integers a and b , subtracting b from a is equivalent to adding the additive inverse of b to a :

$$a - b = a + (-b)$$

Specifically, subtracting a negative integer is equivalent to adding its positive counterpart:

$$a - (-b) = a + b.$$

(i) **6 - 24**

$$6 - 24 = 6 + (-24) = \mathbf{-18}$$

(ii) **18 - (-8)**

$$18 - (-8) = 18 + 8 = \mathbf{26}$$

(iii) **(-16) - (-5)**

$$(-16) - (-5) = -16 + 5 = \mathbf{-11}$$

(iv) **(-20) - 6**

$$(-20) - 6 = -20 + (-6) = \mathbf{-26}$$

(v) **(-1) - (-19)**

$$(-1) - (-19) = -1 + 19 = \mathbf{18}$$

(vi) **8 - (-23)**

$$8 - (-23) = 8 + 23 = \mathbf{31}$$

4. Verify the following

To verify equations, we evaluate the Left-Hand Side (L.H.S.) and Right-Hand Side (R.H.S.) independently.

(i) $(-14) + 9 = 9 + (-14)$

L.H.S. $= -14 + 9 = -5$

R.H.S. $= 9 + (-14) = 9 - 14 = -5$

Since L.H.S. = R.H.S. = -5 , the equation is verified.

Property illustrated: Commutative Law of Addition ($a + b = b + a$).

(ii) $(-8) + (-12) = (-12) + (-8)$

L.H.S. $= -8 + (-12) = -20$

R.H.S. $= -12 + (-8) = -20$

Since L.H.S. = R.H.S. = -20 , the equation is verified.

Property illustrated: Commutative Law of Addition.

(iii) $(-6) + [(-8) + 12] = [(-6) + (-8)] + 12$

L.H.S. $= -6 + [4] = -2$

R.H.S. $= [-14] + 12 = -2$

Since L.H.S. = R.H.S. = -2 , the equation is verified.

Property illustrated: Associative Law of Addition ($a + (b + c) = (a + b) + c$).

(iv) $[(-9) + (-7)] + (-14) = (-9) + [(-7) + (-14)]$

L.H.S. $= [-16] + (-14) = -30$

R.H.S. $= -9 + [-21] = -30$

Since L.H.S. = R.H.S. = -30 , the equation is verified.

Property illustrated: Associative Law of Addition.

5. Fill in the blanks

(i) $8 + \underline{\hspace{1cm}} = 0$

Let the blank be x . $8 + x = 0 \implies x = -8$.

Answer: -8 (Additive Inverse property)

(ii) $(-10) + \underline{\hspace{1cm}} = -10$

Let the blank be x . $-10 + x = -10 \implies x = 0$.

Answer: 0 (Additive Identity property)

(iii) $(-6) + (-8) = (-8) + \underline{\hspace{1cm}}$

By the Commutative Law, $a + b = b + a$.

Answer: -6

(iv) $(-6) + \underline{\hspace{1cm}} = -14$

Let the blank be x . $-6 + x = -14 \implies x = -14 - (-6) = -14 + 6 = -8$.

Answer: -8

(v) $(-11) + \underline{\hspace{1cm}} = -7$

Let the blank be x . $-11 + x = -7 \implies x = -7 - (-11) = -7 + 11 = 4$.

Answer: 4

(vi) $(-9) + \underline{\hspace{1cm}} = -1$

Let the blank be x . $-9 + x = -1 \implies x = -1 - (-9) = -1 + 9 = 8$.

Answer: 8

6. Subtract the sum of -137 and -43 from the sum of -103 and 27

Let S_1 be the sum of -137 and -43 , and S_2 be the sum of -103 and 27 .

$$S_1 = -137 + (-43) = -(137 + 43) = -180$$

$$S_2 = -103 + 27 = -(103 - 27) = -76$$

Subtracting S_1 from S_2 :

$$S_2 - S_1 = -76 - (-180) = -76 + 180 = \mathbf{104}$$

Answer: **104**

7. Subtract -29 from -53 and add -16 to the result

First, subtract -29 from -53 :

$$-53 - (-29) = -53 + 29 = -24$$

Next, add -16 to this result:

$$-24 + (-16) = -(24 + 16) = \mathbf{-40}$$

Answer: **-40**

8. The sum of two integers is 43 . If one of them is -27 , find the other

Let the unknown integer be x .

$$x + (-27) = 43$$

$$x - 27 = 43$$

$$x = 43 + 27 = \mathbf{70}$$

Answer: **70**

9. Find the successor of**Successor of an Integer**

The **successor** of any integer a is defined as $a + 1$ (the immediate integer to its right on the number line).

(i) $\mathbf{30} \implies 30 + 1 = \mathbf{31}$

(ii) $-70 \implies -70 + 1 = \mathbf{-69}$

(iii) $-206 \implies -206 + 1 = \mathbf{-205}$

(iv) $-1 \implies -1 + 1 = \mathbf{0}$

10. Find the predecessor of**Predecessor of an Integer**

The **predecessor** of any integer a is defined as $a - 1$ (the immediate integer to its left on the number line).

(i) $60 \implies 60 - 1 = 59$

(ii) $-351 \implies -351 - 1 = -352$

(iii) $0 \implies 0 - 1 = -1$

(iv) $-99 \implies -99 - 1 = -100$